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 Studierichting: Informatica  
 Oefeningenreeks 6B nr. 21.1

$$\begin{aligned}
 & \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots + \frac{1}{(2n-1)(2n+1)} \\
 s_n &= \frac{1}{2} \left( 1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \dots + \frac{1}{2n-1} - \frac{1}{2n+1} \right) \\
 &= \frac{1}{2} \left( 1 - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{5}} + \cancel{\frac{1}{5}} - \cancel{\frac{1}{7}} + \dots + \cancel{\frac{1}{2n-1}} - \frac{1}{2n+1} \right) \\
 &= \frac{1}{2} \left( 1 - \frac{1}{2n+1} \right) \\
 &= \frac{1}{2} \left( \frac{2n+1}{2n+1} - \frac{1}{2n+1} \right) \\
 &= \frac{1}{2} \cdot \frac{2n}{2n+1} \\
 &= \frac{n}{2n+1}
 \end{aligned}$$

## References